

From the Cover: Caloric restriction improves memory in elderly humans



Animal studies suggest that diets low in calories and rich in unsaturated fatty acids (UFA) are beneficial for cognitive function in age. Here, we tested in a prospective interventional design whether the same effects can be induced in humans. Fifty healthy, normal- to overweight elderly subjects (29 females, mean age 60.5 years, mean body mass index 28 kg/m²) were stratified into 3 groups: (i) caloric restriction (30% reduction), (ii) relative increased intake of UFAs (20% increase, unchanged total fat), and (iii) control. Before and after 3 months of intervention, memory performance was assessed under standardized conditions. We found a significant increase in verbal memory scores after caloric restriction (mean increase 20%; $P < 0.001$), which was correlated with decreases in fasting plasma levels of insulin and high sensitive C-reactive protein, most pronounced in subjects with best adherence to the diet (all r values < -0.8 ; all P values < 0.05). Levels of brain-derived neurotrophic factor remained unchanged. No significant memory changes were observed in the other 2 groups. This interventional trial demonstrates beneficial effects of caloric restriction on memory performance in healthy elderly subjects. Mechanisms underlying this improvement might include higher synaptic plasticity and stimulation of neurofacilitatory pathways in the brain because of improved insulin sensitivity and reduced inflammatory activity. Our study may help to generate novel prevention strategies to maintain cognitive functions into old age.